

A Power Model for D-MIMO Systems: Comparing the Energy Consumption of D-MIMO and Collocated MIMO

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Abstract—TODO
Index Terms—

I. INTRODUCTION

TODO small paragraph on energy consumption of 5G 6G networks

Recent works have focused on characterizing this energy consumption of wireless networks, and in particular of multiple input multiple output (MIMO) systems. In [1], the authors propose a detailed power consumption model for collocated MIMO systems, which is based on the power consumption of the different hardware components in the system for frequency range 3 (FR3) which is envisioned for 6G systems. Another prevelant enabler for 6G systems is distributed MIMO (D-MIMO)/cell-free MIMO. However, works that characterize the power consumption of D-MIMO systems is limited. TODO add som refs and work here Moreover, it is unclear whether the power consumption of D-MIMO systems is is on par, lower or higher than the one of collocated MIMO systems. Especially, considering the additional network consumption that occurs from having multiple access point (AP), fronthaul links, synchronization, etc. In this work we introduce an extension of the power model introduced in [1] for the D-MIMO scenario. We incorporate TODO mention all extensions. TODO mention it is opensource an where it can be found.

II. CONCLUSION

TODO future work:

- include channel estimation error in rate calculation
- include channel estimation power consumption in power model
- include pilot contamination in rate calculation
- make some parameters more realistic e.g. fronthaul power consumption, synchronization power consumption, etc.
- investigate the impact of hardware impairments on system performance
- accurately characterize all parameters based on the size of the BS/APs.

REFERENCES

- [1] E. Peschiera, S. Yun, Y. Lee, L. V. der Perre, and F. Rottenberg, "A parametric power model of upper mid-band (fr3) base stations for 6g," in *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 21 476–21 480.

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